SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES

DEPARTMENT OF FINANCIAL ENGINEERING

PENSION FUND EFFICIENCY TOOLKIT: A PROTOTYPE SYSTEM FOR FRONTIER-BASED OPTIMIZATION ANALYSIS OF SELF-ADMINISTERED PENSION FUNDS IN ZIMBABWE

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Title Page

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Declaration

I the author, LORRAINE MUJURU H220021Y do hereby declare that this work has not previously been accepted in substance for any degree and is not being concurrently submitted in candidature for any degree.

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Dedication

This prototype is dedicated to my family, whose unwavering support throughout my studies at HIT made this work possible. The sacrifices they made and the encouragement they offered during difficult periods are what carried me to the finish line. I hope this achievement brings them the pride they deserve.

Acknowledgements

I would like to extend sincere gratitude to my supervisor for the guidance and mentorship provided throughout the development of this prototype. The constructive feedback at each stage of the project meaningfully shaped both the quality and direction of the final system.

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Lastly, I thank my fellow students and friends at HIT. Four years of shared experiences, late-night study sessions, and diverse perspectives have had a genuine impact on who I am as both a student and a person.

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Abstract

The Pension Fund Efficiency Toolkit is a software prototype that operationalises the analytical framework from the accompanying research study on frontier-based efficiency measurement of Zimbabwean self-administered pension funds. The system translates a multi-stage pipeline — covering data ingestion, efficiency scoring, peer benchmarking, and contextual factor analysis — into an interactive web-based dashboard designed for pension fund managers. Built in Python 3.11 using Streamlit and a set of specialised open-source libraries, the prototype enables users to upload fund data, execute the full analysis pipeline without any programming knowledge, and generate professional PDF reports suitable for board presentations and regulatory submissions. The system demonstrates that rigorous financial modelling can be made accessible to non-technical practitioners through careful interface design without sacrificing analytical integrity.

Abbreviation

Abbreviation	**Full Form**
DEA	Data Envelopment Analysis
CCR	Charnes, Cooper and Rhodes Model
BCC	Banker, Charnes and Cooper Model
CRS	Constant Returns to Scale
VRS	Variable Returns to Scale
IRS	Increasing Returns to Scale
DRS	Decreasing Returns to Scale
RF	Random Forest
PDF	Portable Document Format
UI	User Interface
IPEC	Insurance and Pensions Commission
HIT	Harare Institute of Technology
SQLite	Self-Contained Relational Database Engine
CSV	Comma-Separated Values

Table 1: Glossary of Abbreviations

Chapter 1

INTRODUCTION

Zimbabwe's pension sector has faced considerable turbulence over the past decade. Hyperinflation eroded fund values substantially, exchange rate instability disrupted investment portfolios, and regulatory scrutiny intensified following several high-profile fund failures. Against this backdrop pension fund managers need practical tools — not just academic models — to understand how efficiently their funds are operating and where improvements are realistically achievable.

The Pension Fund Efficiency Toolkit is the software implementation of a research study on efficiency measurement in Zimbabwe's self-administered standalone pension funds. The study applied Data Envelopment Analysis (DEA) to benchmark funds against each other and then used a Random Forest regression model to identify which structural and environmental factors drive the observed efficiency differences. This prototype takes that analytical pipeline from a research setting into practice, packaging it into an interactive dashboard that a fund manager can actually sit down and use. No programming is required.

The system processes panel data from up to twenty registered Zimbabwean pension funds. It produces efficiency scores, identifies top-performing peer funds, generates specific cost-reduction recommendations, and exports formatted PDF reports. All of this happens through a web interface accessible from a standard browser.

1.1 Problem Statement

Pension fund managers in Zimbabwe currently lack accessible tools to systematically evaluate operational efficiency. Regulators like the Insurance and Pensions Commission (IPEC) collect annual financial disclosures, but there is no standardised mechanism for funds to benchmark themselves against peers or receive spe-

cific, data-driven guidance on improving performance. Those who want to understand their fund's relative position typically rely on manual spreadsheet comparisons — time-consuming work that is prone to error and fails to capture the multi-dimensional nature of fund performance.

The analytical framework that addresses these limitations — Data Envelopment Analysis — computes a frontier of best-practice performance across the peer group and measures each fund's distance from that frontier. DEA and its extensions have historically been the domain of researchers. The gap between what the literature makes available and what fund managers can practically access is exactly what motivated the development of this system.

1.2 System Objectives

The Pension Fund Efficiency Toolkit was designed to achieve the following objectives:

- To implement a fund efficiency scoring pipeline that evaluates both overall and internal operational efficiency for up to twenty Zimbabwean pension funds
- To automatically identify best-practice benchmark funds and calculate specific, actionable input-reduction targets for underperforming funds in dollar terms
- To determine which structural and environmental factors most strongly influence fund efficiency across the sector as a whole
- To deliver an interactive browser-based dashboard requiring no technical expertise to operate, so that fund managers can run the full analysis independently
- To persist uploaded fund data in a local database so that managers do not need to re-upload files on subsequent visits
- To generate downloadable PDF reports summarising findings for stakeholders and regulatory purposes

Conclusion

This chapter established the sector context and motivation behind the prototype. Zimbabwe's pension funds operate in a difficult environment and evidence-based efficiency measurement can help managers make better decisions with the resources

they have. The objectives show that the system aims to bridge the gap between research-grade analytical methods and practical day-to-day fund management. The chapters that follow document the development process, feasibility considerations, and the full system implementation.

Chapter 2

PROCESS MODELLING

2.1 Introduction

Building this prototype meant choosing a development approach that could accommodate both the technical complexity of the underlying analytical models and the requirement that the resulting interface be simple for non-technical users to operate. This chapter covers the process model chosen, a feasibility analysis across four dimensions, and the security architecture that governs system access.

2.2 Process Model (Iterative)

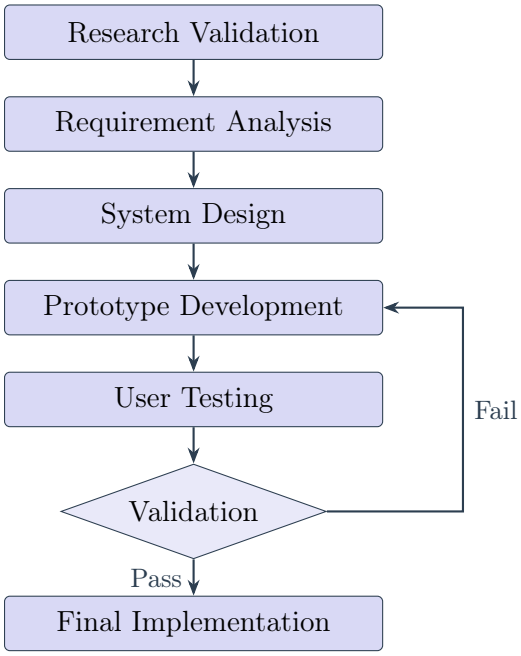

Figure 2.1: Process Model (Iterative)

The iterative prototyping model was selected for this project for several reasons. Financial analysis software of this kind depends on tight feedback loops between the analytical engine and the interface. Getting the DEA pipeline to produce correct results is one thing; surfacing those results in a way that a fund manager actually finds interpretable is quite another. Iterative development allowed both concerns to be addressed in cycles rather than requiring a complete design commitment upfront.

The approach also suited the nature of the underlying algorithms. The bootstrap bias-correction module and the Random Forest second-stage analysis each required calibration iterations. Being able to revisit the design of these components after observing their outputs in the dashboard context — rather than validating them in isolation — produced a more coherent final product. Each cycle of the model above maps roughly to one development sprint, with the validation step corresponding to functional testing of the dashboard against expected analytical outputs.

2.3 Feasibility Analysis

- **Economic Feasibility**: The entire system is built on open-source technologies. Python, Streamlit, PuLP, scikit-learn, and SQLite carry no licensing costs. Compared to commercial financial analytics platforms that charge per-seat fees running into thousands of dollars annually, this prototype achieves equivalent functionality at essentially zero cost while being tailored specifically to the Zimbabwean pension context.
- **Technical Feasibility**: The DEA linear programmes solve in milliseconds per fund using the PuLP/CBC solver on standard hardware. The bootstrap procedure — which runs two hundred or more resampling iterations to produce statistically valid efficiency estimates — completes in under a minute for nineteen funds on a modern laptop. Streamlit handles the web interface without requiring a dedicated server; a single command launches the application locally, making deployment straightforward even where IT infrastructure is limited.
- **Social Feasibility**: Pension fund managers are not, in general, quantitative analysts. The dashboard design reflects this reality. Efficiency scores are expressed as percentages with plain-English status labels (Excellent, Good, Fair, Needs Attention), recommendations are stated as specific dollar amounts rather than technical slack values, and academic terminology is kept away from the interface entirely. The system was refined iteratively

to ensure that outputs are interpretable without any background in DEA or statistics.

- **Operational Feasibility**: The system accepts CSV files in a format consistent with data that fund managers already compile for IPEC submissions. No infrastructure changes are required on the user's side. The built-in SQLite database means that uploaded data persists between sessions, so a manager does not need to re-upload the same dataset each time they log in. The application runs locally on a Windows or Linux machine with Python installed.

2.4 System Security

Access to the system is controlled by credential-based login. The administrator account representing the fund manager is configured through environment variables rather than being stored in the database, which means the primary credentials are never written to disk in plain text form. The password is managed solely by the system developer and the designated user.

The application does not expose any network endpoints beyond the local machine by default. All fund data is stored in a local SQLite file rather than on a remote server, reducing the attack surface considerably. The security model is intentionally simple and proportionate to the single-user deployment context: there is one authenticated user, and unauthenticated sessions are immediately redirected to the login page.

Conclusion

Through iterative development the prototype successfully addressed the gap between research-grade analysis and practical day-to-day use. The feasibility study confirms that the system is economically viable, technically sound, socially appropriate for its intended users, and operationally compatible with existing workflows in Zimbabwean pension fund management. The security architecture is straightforward and appropriate for the deployment context.

Chapter 3

SOFTWARE METHODOLOGY

3.1 Introduction

This chapter describes the implementation of the Pension Fund Efficiency Toolkit at the system level. It covers the overall flow of the application from the user's perspective, describes each screen and its purpose, and provides representative code extracts from the core modules. The aim is to give a clear picture of how the system works from the front end all the way through to the data layer.

3.2 System Flowchart

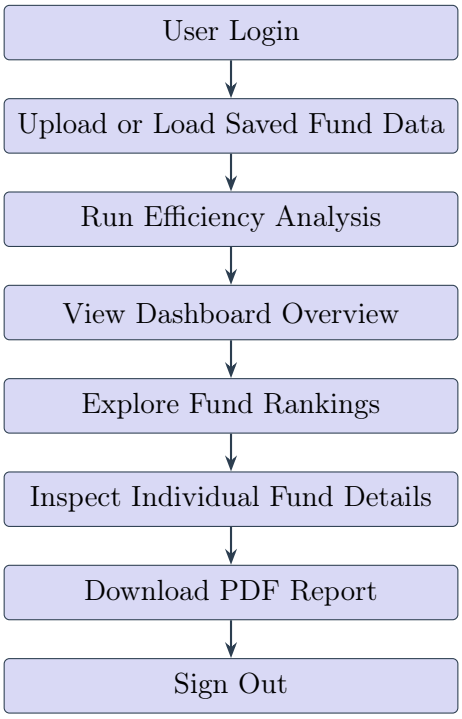

Figure 3.1: System Flowchart

The flowchart maps the primary use path through the application. When a user authenticates, the system checks whether they have any previously uploaded datasets saved in the local database and loads the most recent one automatically — so returning users land directly on the dashboard without needing to re-upload anything. If no saved data exists, the CSV upload widget in the sidebar is used. Clicking Run Analysis triggers the full pipeline: CCR efficiency scoring, BCC efficiency scoring, scale efficiency classification, bootstrap bias correction, and the Random Forest second-stage determinant analysis. Results are stored in the session and the tabbed dashboard allows exploration at whatever level of detail the manager needs.

3.3 System Description

3.3.1 Step 1: Login

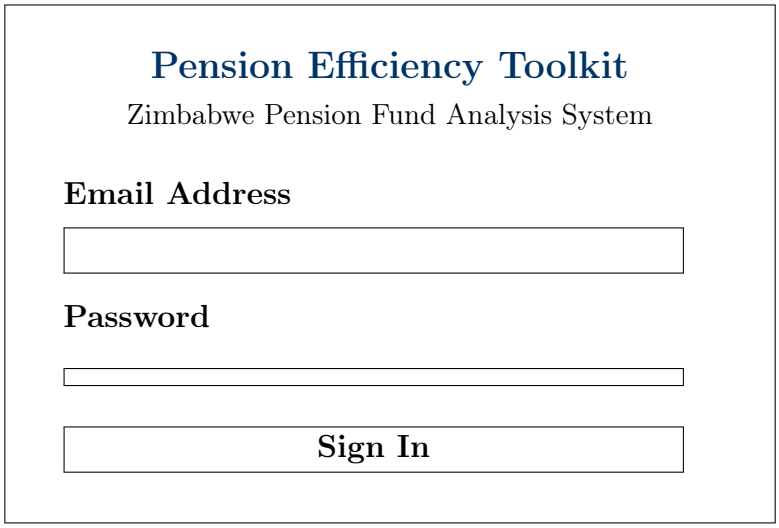

Figure 3.2: Login Screen

The login screen is the system's entry point. The user enters their email address and password. Authentication is handled against credentials stored in environment variables on the server side — the fund manager's email and password are set at deployment time and are never held in the database. Failed attempts return a clear error message; successful login redirects to the main dashboard immediately.

3.3.2 Launch Page — Dashboard Overview

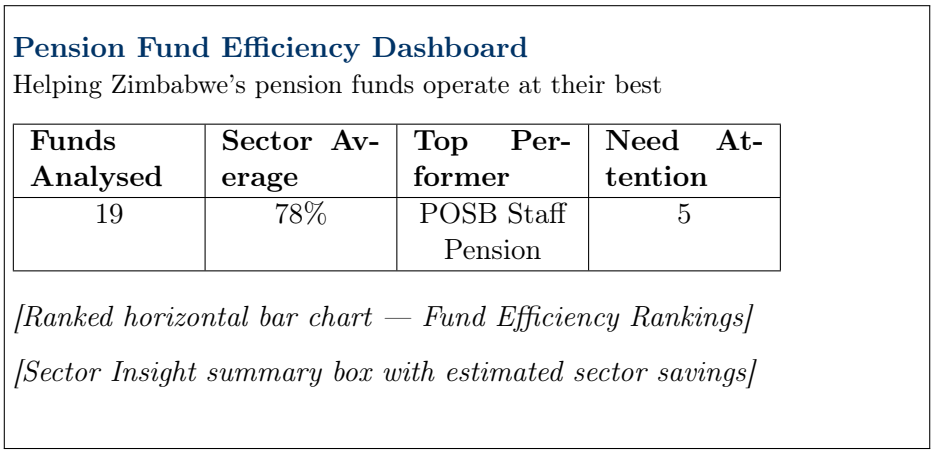

Figure 3.3: Dashboard Overview

After login the user arrives at the main dashboard. The sidebar shows any previously saved datasets and allows a new CSV upload. Once data is loaded and Run Analysis is clicked, the overview tab displays four headline metrics: number of funds analysed, sector average efficiency, the top-performing fund, and how many funds fall below the 75 % threshold. A ranked horizontal bar chart shows all funds by efficiency score, colour-coded from red through orange to green. At the bottom a plain-language insight box summarises the sector picture and estimates total annual savings available if underperforming funds reached the efficiency frontier.

3.3.3 Fund Rankings

Fund Performance Rankings

Every fund is scored on how efficiently it converts resources into returns for members. A score of 100% means the fund is on the frontier.

#	**Fund Name**	**Score**	**Status**	**Savings**
1	POSB Staff Pension Fund	100.0%	Excellent	—
2	Zimplats Mineworkers Pension	98.3%	Excellent	—
3	Delta Corporation Pension	91.5%	Excellent	$1.2M
...	...	...	...	...

[Scatter plot: Overall Efficiency vs. Internal Efficiency]

Figure 3.4: Fund Rankings Tab

The rankings tab presents a full table of all funds ordered by efficiency score. Each row shows rank, fund name, score, status label, and estimated annual savings potential. Below the table a scatter plot compares overall efficiency against internal operational efficiency. This helps distinguish funds that are simply operating at the wrong scale from those with genuine operational inefficiencies. Funds sitting on the diagonal line have matching scores across both measures.

3.3.4 Fund Details

Select a Fund to View CBZ Bank Staff Pension Fund ∨

[Gauge: 68%]

CBZ Bank Staff Pension Fund

Ranked 12 of 19 funds

Efficiency Score: 68.0% | Sector Average: 78%
Gap to Average: −10.0%

This fund is underperforming relative to its peers.

Recommended Actions:

Reduce Operating Expenses: \$3.4M → \$2.1M (saves \$1.3M/year)

Learn From: POSB Staff Pension Fund | Zimplats Mineworkers Pension Fund

Figure 3.5: Fund Details Tab

This tab lets the manager select any individual fund and drill into its performance. A gauge chart shows the efficiency score visually, with colour zones running from red at the bottom to green at the top. To the right the fund's rank, score relative to the sector average, and a plain-language interpretation are displayed. Below the gauge, specific recommendations list each input where a reduction is achievable, showing the current amount, the recommended target, and the estimated annual saving. A peer benchmarking section at the bottom identifies the most similar high-performing funds so the manager can investigate what those funds do differently in practice.

3.3.5 Download Report

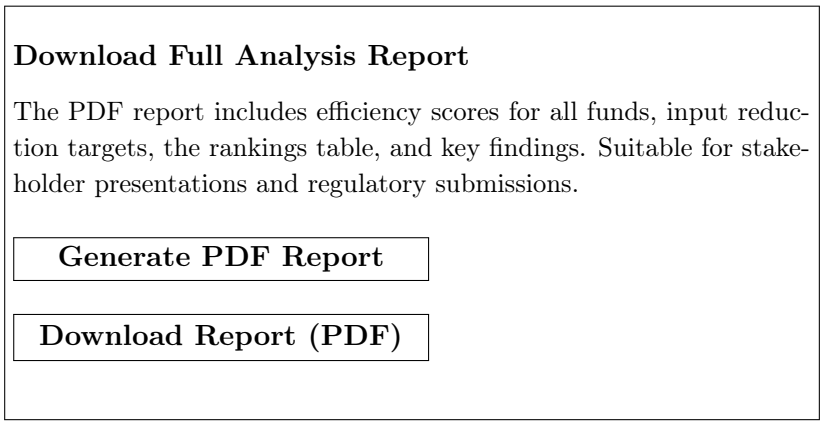

Figure 3.6: Download Report Tab

The reports tab provides a single button that generates a formatted PDF report using ReportLab. The document includes an executive summary, the complete efficiency rankings table, scale efficiency classifications, input reduction targets for each fund, and a feature importance chart showing which factors most influence fund efficiency across the sector. Once generated, a download button appears so the file can be saved locally.

Conclusion

The implemented methodology translates the research's analytical framework into working operational components, with each computational stage traceable to specific methods and algorithms in the accompanying study document. The system architecture ensures all calculations maintain analytical rigour while the user interface abstracts that complexity into plain-language outputs that a fund manager can act on directly. The modular code structure makes the system straightforward to maintain and extend as the dataset grows or as future research refines the underlying methods.

APPENDIX

3.3.6 System Code

The authentication module verifies the fund manager's credentials against environment variables. Because credentials are never stored in the database, there is no risk of plaintext password exposure in the SQLite file.

Listing 3.1: Login credential verification — auth.py

```python
def login(email: str, password: str) -> tuple[bool, str, bool, int]:
    """Attempt login. Returns (success, display_name, is_admin, user_id)."""
    from pension_toolkit.db import ADMIN_USER_ID, get_user_by_email

    email = email.strip()

    # Admin credentials come from environment variables, never the database
    if is_admin_email(email):
        if password == _admin_password():
            return True, _admin_name(), True, ADMIN_USER_ID
        return False, "", False, -1

    user = get_user_by_email(email)
    if user and user["password"] == password:
        return True, user["full_name"], False, user["id"]
    return False, "", False, -1
```

The pipeline function aggregates multi-year panel data to a cross-sectional representation and then orchestrates the DEA solver, scale analysis, bootstrap procedure, and Random Forest second stage in sequence.

Listing 3.2: Analysis pipeline — ui_streamlit.py

```python
def run_pipeline(df: pd.DataFrame, B: int, seed: int) -> dict:
    numeric_cols = df.select_dtypes(include="number").columns.tolist()
    df_agg = df.groupby("fund_id")[numeric_cols].mean().reset_index()
    fund_name_map = df.groupby("fund_id")["fund_name"].agg(
        lambda x: x.mode()[0])
    df_agg = df_agg.merge(fund_name_map, on="fund_id")

```

```
    X_in, Y_out, fund_ids = get_dea_matrices(df_agg, INPUT_COLS,
        OUTPUT_COLS)
    ccr   = dea_ccr_input_oriented(X_in, Y_out, fund_ids)
    bcc   = dea_bcc_input_oriented(X_in, Y_out, fund_ids)
    scale = compute_scale_efficiency(ccr, bcc)

    def _ccr_func(X, Y):
        return dea_ccr_input_oriented(X, Y, fund_ids)

    boot = simar_wilson(_ccr_func, X_in, Y_out,
                        fund_ids=fund_ids, B=B, seed=seed)

    ...
    return dict(ccr=ccr, bcc=bcc, scale_df=scale_df,
                boot_df=boot_df, rf_result=rf_result, ...)
```

The data persistence layer stores every CSV upload with its fund data rows so that results survive browser restarts. The upload history table allows the manager to switch between multiple datasets without re-uploading.

Listing 3.3: Fund data persistence — db.py

```
def save_fund_data(df: pd.DataFrame, filename: str, user_id: int)
    -> int:
    """Persist a new upload for the given user. Returns the new
        upload_id."""
    with _connect() as conn:
        _init_tables(conn)
        n_rows  = len(df)
        n_funds = int(df["fund_id"].nunique())
        uploaded_at = datetime.now().isoformat(timespec="seconds")

        cur = conn.execute(
            "INSERT INTO upload_history "
            "(user_id, filename, uploaded_at, n_rows, n_funds) "
            "VALUES (?, ?, ?, ?, ?)",
            (user_id, filename, uploaded_at, n_rows, n_funds),
        )
        upload_id = cur.lastrowid
        df_to_save = df.copy()
        df_to_save["upload_id"] = upload_id
        df_to_save.to_sql("fund_data", conn,
                          if_exists="append", index=False)
        return upload_id
```